E-mail: rayssa.feitosa@alumni.usp.br

Prioritization of Corregulatory miR-mRNA modules (PCmiRm): an innovative tool for comparing groups using miRNA-mRNA coregulation modules based on miRNA differential expression and tissue-specific targets.

Feitosa, Rayssa M.M.W.[1]; Paradiso, Jana A. O.[2]; Machado-Lima, Ariane[2]; Brentani, Helena[1].

[1] Psychiatry Institute of the Faculty of Medicine Clinics Hospital - University of Sao Paulo, Sao Paulo, Brazil
[2] The School of Arts, Sciences and Humanities (EACH) - University of Sao Paulo, Sao Paulo, Brazil

Background and Proposal

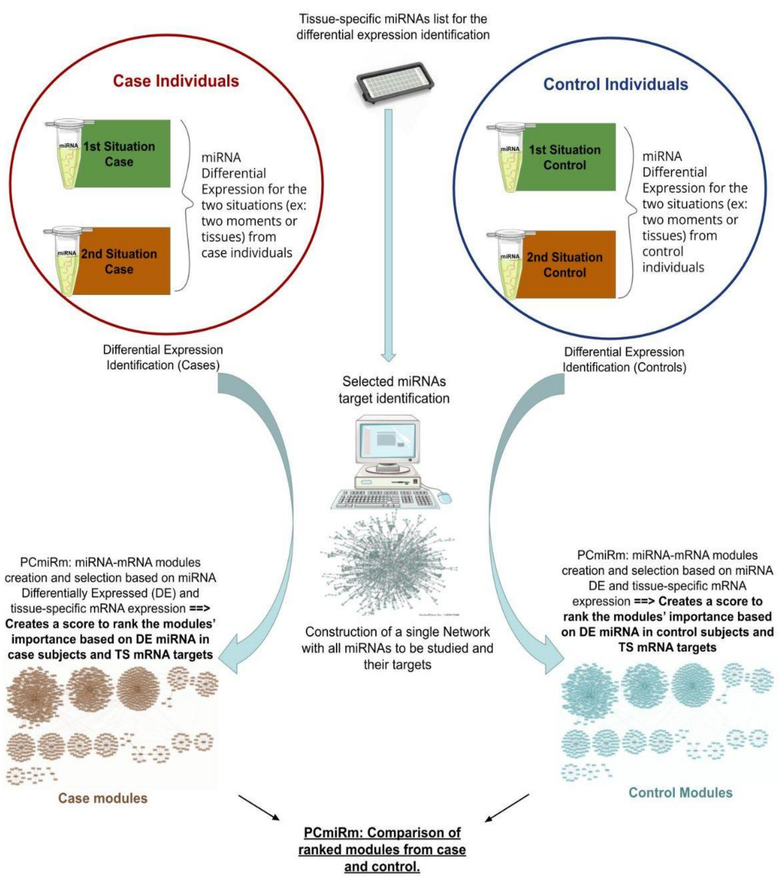

Fig. 1: Study proposal Scheme. DE = Differentially Expressed.

Frequently, to analyze the natural organic effect of miRNAs, miRNA-target networks are created. The existing tools used to analyze miRNA-target networks do not consider miRNA differential expression in an individual through time when comparing two groups. Additionally, in cases where you do not use mRNA expression data, the targets are computationally identified, which unfortunately leads to nonspecific targets.

Here, we proposed a new tool to analyze miRNA-mRNA networks that measure miRNA differential expression in two instants of time, for two different groups, a frequent situation found in clinical studies. In order to do so, PCmiRm needs 4 miRNA inputs (up and down-regulated miRNAs comparing two instances of time for each group of study - for example, cases versus control).

Methods

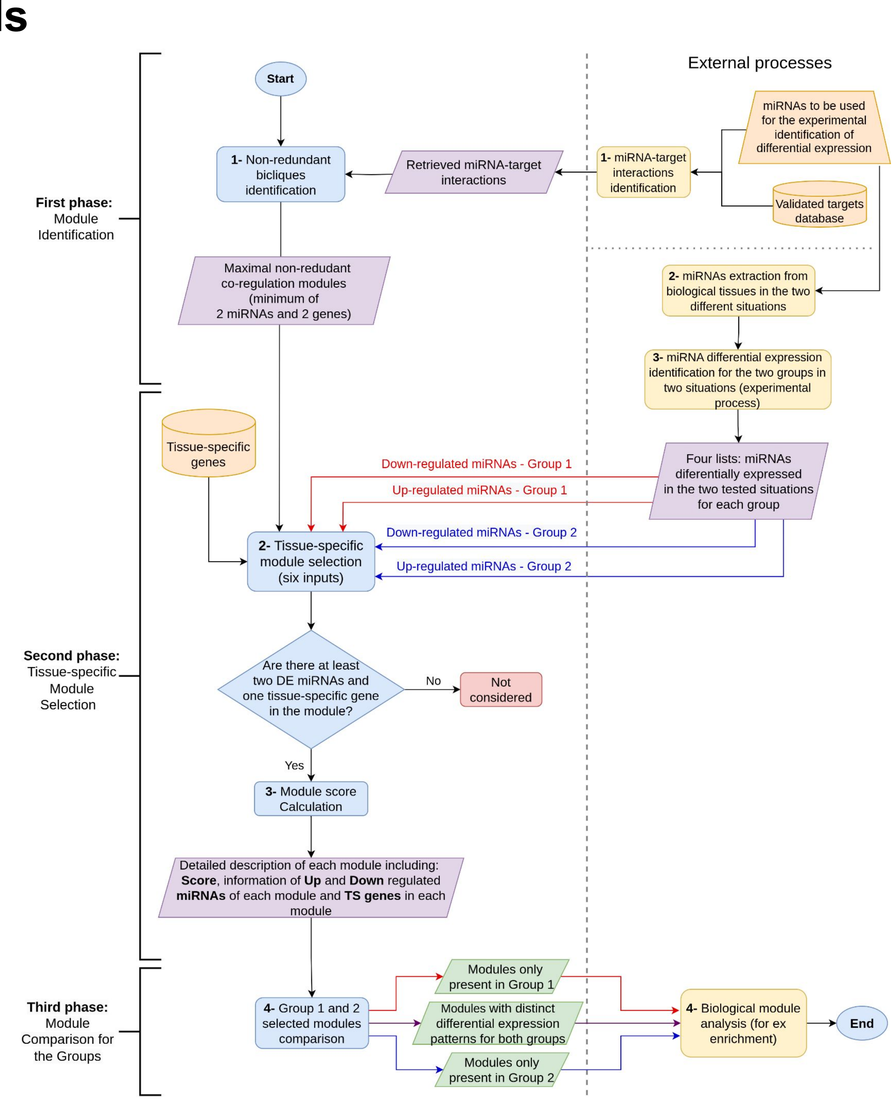

Fig. 2: Methodology Flowchart.

First Phase - Module Creation

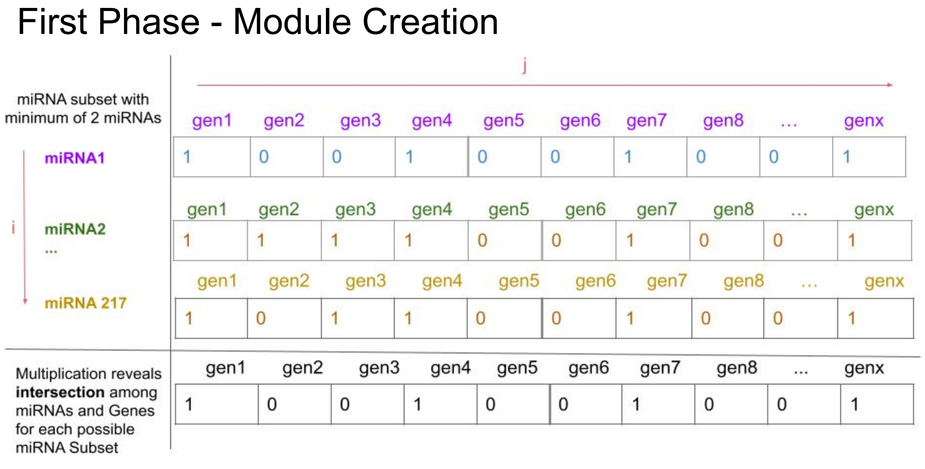

Fig. 3: Matrix for modules creation.

PCmiRm exhausts the possibilities of interactions between a group of miRNAs with a group of mRNAs (at least 2 of each) creating multiple miRNA-mRNA modules as shown in Fig 3.

Second Phase - Module Selection

Selection of the modules containing at least two DE miRNAs and one Tissue-Specific gene followed by ranking using the formula:

$$Ms = \frac{(miup + midw)}{Tmi} \; x \; (log_2(Tmi) \; x \; miW) \; + \; \frac{Sg}{Tg} \; x \; (log_2(Tg) \; x \; SgW)$$

Third Phase - Module Comparison for Each Group

In this Phase the different and equal modules are identified. The user has the option to choose all modules or those with the with the 5, 10, 20, and 50% highest scores to compare. Two modules are considered equal when they have exactly the same miRNAs and target genes, and each miRNA has the same differential expression pattern (up-regulated or down-regulated) in both modules.

Case Study Results:

We used 217 literature-curated placental miRNAs that presented 10 142 interactions with 3 328 genes after the target identification. This interaction list was used as input for **Phase 1** that resulted in **359 274 597** modules. Considering modules with at least two targets, there were **198 358** modules, and after removing the redundancy, **11 813 maximal modules remained containing 2-281 genes and 2-14 miRNAs**.
To test the behavior of our tool under multiple different situations, we simulated 6 600 miRNA differential expression combinations. For **Phase 2** the target genes and miRNAs were prioritized using miRNA differential expression and Placenta-Specific genes from the database Tissues 2.0.

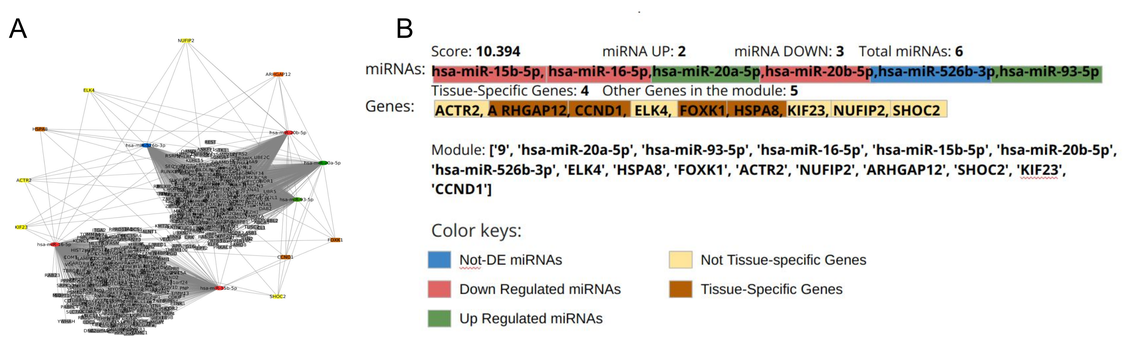

Fig. 4: Modules representation. A: corregulation module (bicliques) represented in the network. B: PCmiRm output.

For **Phase 3** in each of the 6600 hypothetical situation tested, it was obtained from zero to 3247 different modules for cases and controls depending on the modularity of the miRNAs involved. From those the user can select the desired percentage of high scores.

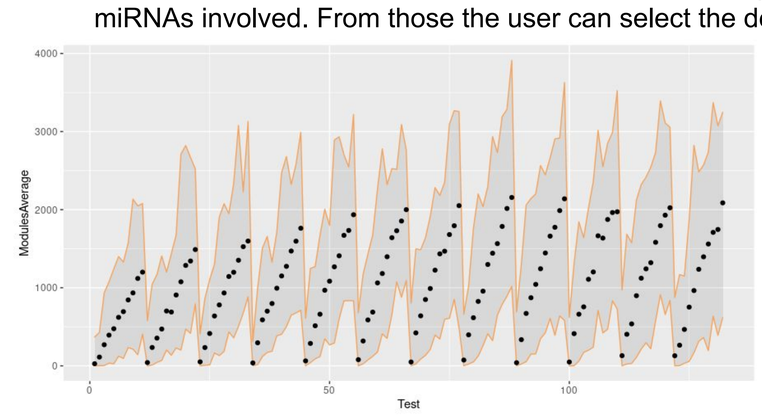

Fig. 5: average number of modules (y-axis) (each dot), and the maximum and minimum number of modules (yellowish limits), for each of the 132 miRNA hypothetical amount of DE (x-axis - "Test").

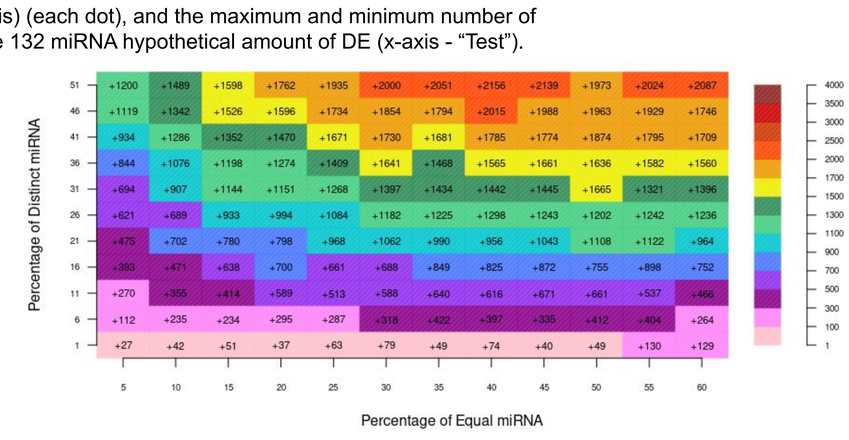

Fig. 6: Average number of modules for each percentage of equally and distinctly DE miRNAs.

Discussion and Conclusion

Our study proposal allowed the creation of a new computational tool that solves two of the problems found in the literature when analyzing biological miRNA-mRNA networks: tissue-specific non-experimental target identification; and the analysis and comparison of bipartite miRNA-mRNA networks considering the biological properties of miRNA activity.

Our test input data showed that PCmiRm can be used for several different situations involving miRNA expression as it identifies different modules for two groups even if there are a small amount of DE miRNAs, depending on the modularization rate of that miRNA. We could also testify that miRNAs rarely act alone since 91% of our miRNAs formed modules with at least two genes, a piece of valuable information that can further help in the miRNA functional studies.